

Trialbank
Independent Trial Paper B

2026

Higher School Certificate

Year 12 Trial HSC Examination

Software Engineering

Total Marks: 80

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours 20 minutes
- The HSC exam is completed online; answer on screen in Trialbank, or in the spaces provided on this paper
- Python is used for code questions
- Write your name at the top of every page

Section I
Total marks (20)

- Attempt Questions 1–16
- Allow about 25 minutes for this section

Section II
Total marks (60)

- Attempt ALL questions
- Allow about 1 hour and 55 minutes for this section

STUDENT NAME: _____

Section I

20 marks

Attempt Questions 1–16

Allow about 25 minutes for this section

Questions 1–8: select ONE answer. Questions 9–12: select ALL correct answers (all must be correct for the mark). Questions 13–16: match each item to ONE option; options may be used more than once.

1 **Which of these is a *non-functional* requirement for a school room-booking app?**

- ◯ **A.** Students can cancel a booking
- ◯ **B.** Every page loads in under 2 seconds
- ◯ **C.** Admins can export bookings to a spreadsheet
- ◯ **D.** Teachers can see who booked a room

2 **A website stores passwords as SHA-256 hashes without a salt. Which attack does this make much easier?**

- ◯ **A.** A rainbow table attack
- ◯ **B.** Cross-site request forgery
- ◯ **C.** A denial-of-service attack
- ◯ **D.** Clickjacking

3 **Which HTTP response header tells a browser to connect to the site only over HTTPS in future?**

- ◯ **A.** Content-Type
- ◯ **B.** Strict-Transport-Security
- ◯ **C.** Cache-Control
- ◯ **D.** Set-Cookie

4 **What is the output of this code?**

```
def f(n):
    if n <= 1:
        return n
    return f(n - 1) + f(n - 2)

print(f(6))
```

- ◯ **A.** 5
- ◯ **B.** 8
- ◯ **C.** 13
- ◯ **D.** 6

5 **Why is a dataset split into training data and testing data?**

- ◯ **A.** To make training faster
- ◯ **B.** To estimate how well the model performs on data it has not seen
- ◯ **C.** To reduce the number of features
- ◯ **D.** To label the data automatically

6 **Which CSS selector targets elements with the class `card` that are inside a `<section>`?**

- ◯ **A.** `section.card`
- ◯ **B.** `section .card`
- ◯ **C.** `.card section`
- ◯ **D.** `section > card`

7 **A login form says "Incorrect password" when the username exists but "User not found" when it doesn't. What vulnerability is this?**

- ◯ **A.** User enumeration (information disclosure)
- ◯ **B.** Cross-site scripting
- ◯ **C.** Buffer overflow
- ◯ **D.** Race condition

8 **Which is the best example of MLOps practice?**

- ◯ **A.** Deploying a new model by hand once a year
- ◯ **B.** Automatically retraining, testing and deploying models, then monitoring them in use
- ◯ **C.** Writing all documentation after the project ends
- ◯ **D.** Testing a model only after customers report problems

9 **Which of these are features of a RESTful API? Select ALL that apply.**

- ☐ **A.** Each request is stateless
- ☐ **B.** Resources are identified by URLs
- ☐ **C.** It requires WebSockets
- ☐ **D.** It uses HTTP methods such as GET and POST
- ☐ **E.** Responses are always HTML pages
- ☐ **F.** Responses are commonly JSON

10 **Select ALL statements that are true for this code.**

```
class Vehicle:
    wheels = 4

    def __init__(self, make):
        self.make = make

    def describe(self):
        return f"{self.make} with {self.wheels} wheels"

class Motorbike(Vehicle):
    wheels = 2

bike = Motorbike("Yamaha")
car = Vehicle("Toyota")
```

- ☐ **A.** bike.describe() returns "Yamaha with 2 wheels"
- ☐ **B.** car.describe() returns "Toyota with 4 wheels"
- ☐ **C.** Motorbike must define its own __init__ before objects can be created
- ☐ **D.** wheels is a class attribute
- ☐ **E.** Motorbike overrides the describe() method
- ☐ **F.** Motorbike inherits __init__ from Vehicle

11 **Which of these are ethical concerns with facial recognition systems? Select ALL that apply.**

- ☐ **A.** Lower accuracy for some demographic groups
- ☐ **B.** Collecting biometric data without consent
- ☐ **C.** Faster image loading
- ☐ **D.** Enabling mass surveillance
- ☐ **E.** Lower electricity use

12 **Which are appropriate ways to store users' passwords? Select ALL that apply.**

- ☐ **A.** In plain text
- ☐ **B.** Hashed with bcrypt
- ☐ **C.** Hashed with Argon2
- ☐ **D.** Encoded in base64
- ☐ **E.** Hashed with MD5
- ☐ **F.** Hashed with PBKDF2 using many iterations

13 **Match each type of testing to its description.** **2 marks**

Item	Answer	Options
1. Unit testing		A. The whole system is tested against the requirements
2. Integration testing		B. The client confirms the system meets their needs
3. System testing		C. A single function or module is tested on its own
4. Acceptance testing		D. Modules are tested to check they work together

14 **Match each security principle to its description.** **2 marks**

Item	Answer	Options
1. Principle of least privilege		A. Several layers of security, so one failure isn't fatal
2. Defence in depth		B. Checking data is the right type, length and format before using it
3. Input validation		C. Users and processes get only the access they need
4. Session management		D. Stored data is unreadable if a disk or database is stolen
5. Encryption at rest		E. Creating, tracking and expiring logged-in sessions securely

15 **Match each technology to its main role in a web app.** **2 marks**

Item	Answer	Options
1. HTML		A. Querying and updating the database
2. CSS		B. The structure and content of a page
3. JavaScript (in the browser)		C. Server-side routes and logic
4. SQL		D. Presentation and layout
5. Python with Flask		E. Interactivity in the browser

16 **Match each machine learning term to its meaning.** **2 marks**

Item	Answer	Options
1. Feature		A. One complete pass through the training data
2. Label		B. An input variable used to make a prediction
3. Epoch		C. The known output the model learns to predict
4. Learning rate		D. How big each weight update is
5. Overfitting		E. A non-linear function applied to a neuron's output
6. Activation function		F. Scoring well on training data but poorly on new data

Section II

60 marks

Attempt ALL questions

Allow about 1 hour and 55 minutes for this section

Answer each question in the spaces provided. Write code in Python unless a question says pseudocode is acceptable.

17 Outline the purpose of adding a *salt* before hashing a password. **2 marks**

18 Explain how a cross-site request forgery (CSRF) attack works, and describe ONE way to prevent it. **3 marks**

19 This Flask route shows search results. **4 marks**

```
@app.route("/search")
def search():
    q = request.args.get("q", "")
    return f"<h1>Results for {q}</h1>"
```

(a) Identify the vulnerability and explain how an attacker could exploit it. **(2 marks)**

(b) Rewrite the route so it is secure. **(2 marks)**

20 A student opens the profile page of a web app. Describe how the browser, the web server and the database work together to display the page. **4 marks**

21 Compare progressive web apps (PWAs) with native mobile apps. **4 marks**

22 `results` is a list of `(name, mark)` tuples. Write a Python function `top_students(results, n)` that returns a list of the names of the `n` highest-scoring students, highest mark first. Students with equal marks are listed alphabetically. If there are fewer than `n` students, return all of them. **5 marks**

For example, `top_students([("Ava", 80), ("Ben", 95), ("Cy", 80)], 2)` returns `["Ben", "Ava"]`.

23 A database has these tables: **3 marks**

```
Students(id, name, year)
Bookings(id, student_id, room, day)
```

Write an SQL query that shows each room and how many times Year 11 students booked it, including only rooms booked more than 5 times, most-booked first.

24 Explain why developers use branches in version control. **2 marks**

25 A school uses a decision tree to predict whether a student will pass, based on attendance and homework completion. **4 marks**

(a) Explain how a decision tree makes a prediction for a new student. **(2 marks)**

(b) Give ONE advantage and ONE disadvantage of using a decision tree instead of a neural network for this task. **(2 marks)**

26 Explain how bias can enter a machine learning system at different stages of its development and use. **4 marks**

27 A spam filter is tested on 100 emails: 40 spam correctly flagged, 10 normal emails wrongly flagged as spam, 5 spam emails missed and 45 normal emails correctly allowed. Calculate the filter's accuracy and its precision. **2 marks**

28 Write Python code for these two classes. **5 marks**

- **BankAccount** has a *private* balance starting at 0, and methods `deposit(amount)`, `withdraw(amount)` and `get_balance()`. Both `deposit` and `withdraw` raise a `ValueError` if the amount is not positive, and `withdraw` also raises one if the amount is more than the balance.
- **SavingsAccount** is a BankAccount created with an `interest_rate` (for example 0.05). Its method `add_interest()` deposits interest on the current balance.

29 With reference to your code in the previous question, explain *encapsulation*, and why the balance should be private. **3 marks**

30 Describe how a development team could use a Kanban board and daily stand-up meetings to manage a software project. **4 marks**

31 A developer is building an app that tracks students' locations so parents can see when their child arrives at school. Discuss the social, ethical and legal issues the developer must consider. **5 marks**

32 Design a secure login system for a web app. Describe the process from a user registering to their authenticated session, justifying the security measure used at each step. **6 marks**

End of paper